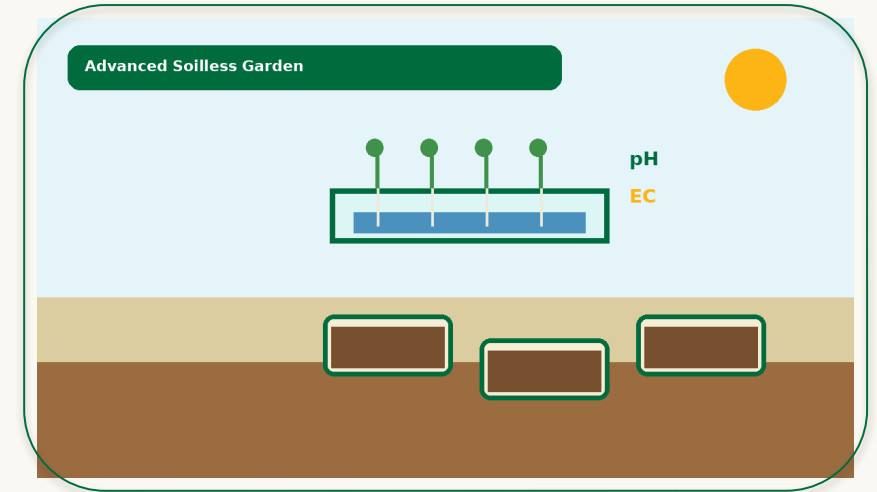


Advanced Soiless Garden

Controlled Soiless Crop Production

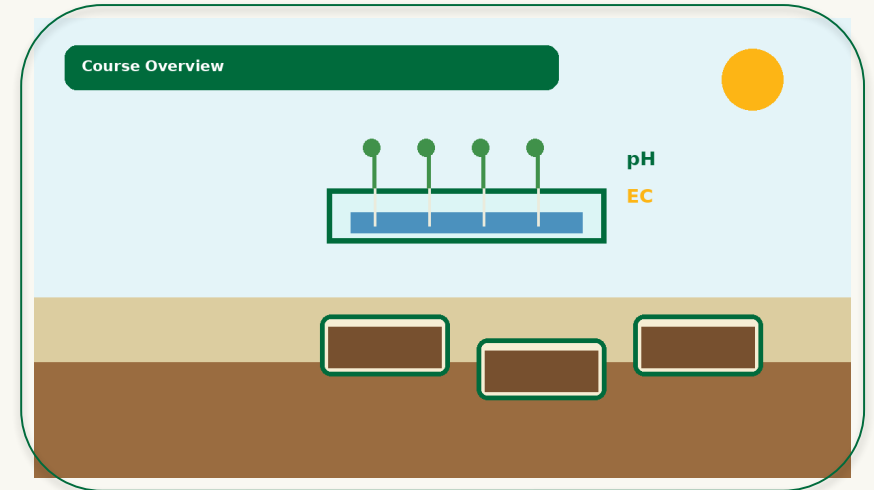
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide quantitative analysis; distinguish measurement from inference; connect evidence to management, engineering and research decisions.
- Core materials: teacher guide, system drawings, approved tools, low-voltage educational hardware where relevant, measuring instruments, meters/sensors, data sheets/logger, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit an engineering design, monitoring plan, experimental protocol or systems-analysis report appropriate to the course.



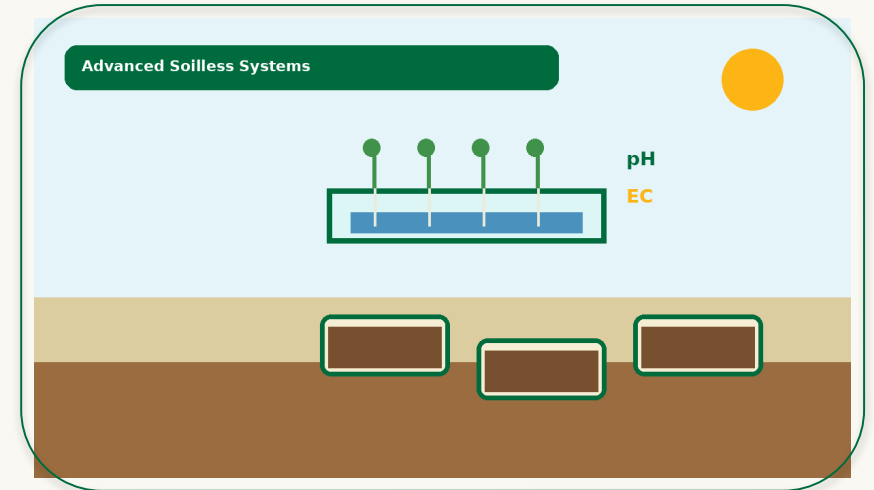
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Advanced Soiless Systems

Teacher technical knowledge, laboratory/field practice and student supervision

- Compare substrate-based, nutrient-film, deep-water and other controlled soiless systems at an introductory engineering level.
- Identify how each system manages root support, nutrient delivery, oxygenation, drainage and water reuse.
- Select system type according to crop, available infrastructure, monitoring capacity and educational objective.
- Distinguish passive from actively pumped systems.
- Keep pumps, electrical equipment and chemical concentrates under authorized adult control.



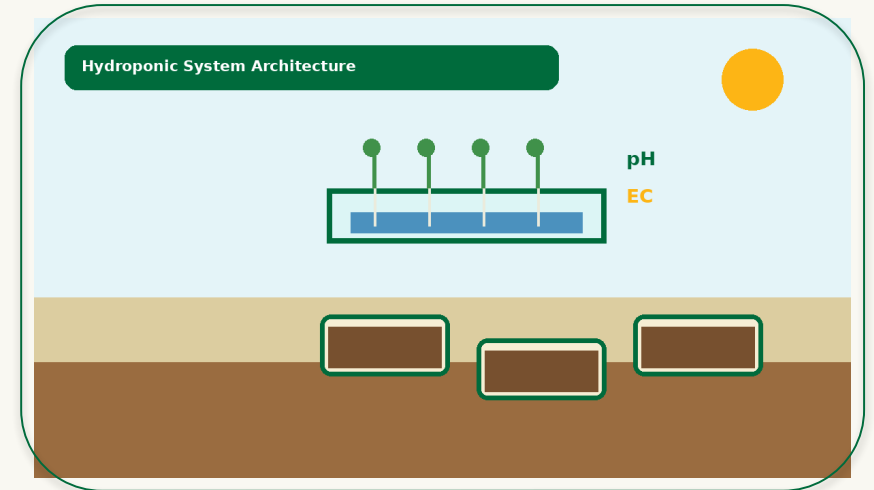
TEACHER PRACTICE

- Teacher task: compare two soiless architectures using common engineering criteria.

Hydroponic System Architecture

Teacher technical knowledge, laboratory/field practice and student supervision

- Map reservoir, pump, delivery line, growing channels/containers, root zone, return line and monitoring points.
- Trace water and nutrient-solution movement through the system.
- Identify failure points such as pump loss, blocked lines, leaks and poor return flow.
- Include isolation, drainage and safe maintenance access in the schematic.
- Document whether the system is open or recirculating.



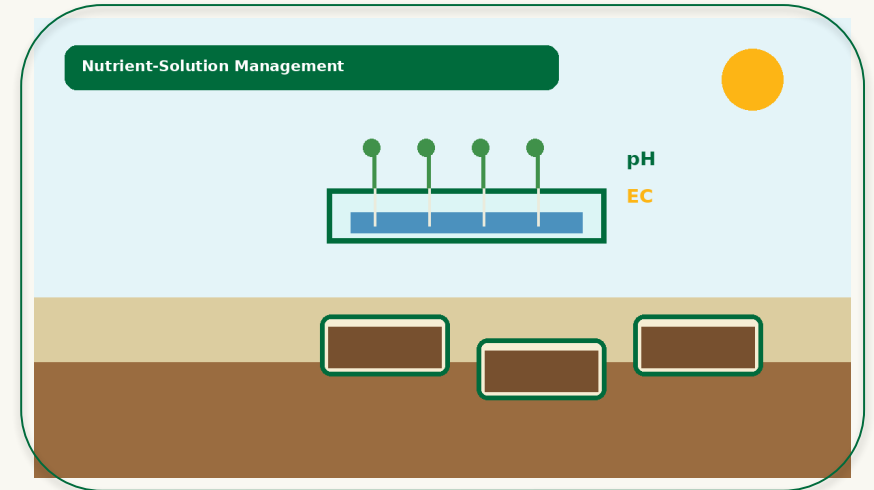
TEACHER PRACTICE

- Produce a complete hydroponic process-flow diagram.

Nutrient-Solution Management

Teacher technical knowledge, laboratory/field practice and student supervision

- Prepare nutrient solutions only from approved formulations and crop-specific guidance.
- Calculate reservoir volume and required dilution using teacher-controlled procedures.
- Record source water, formulation, preparation date, volume and adjustments.
- Monitor solution condition over time rather than assuming the initial mix remains unchanged.
- Keep concentrated nutrient chemicals inaccessible to students.



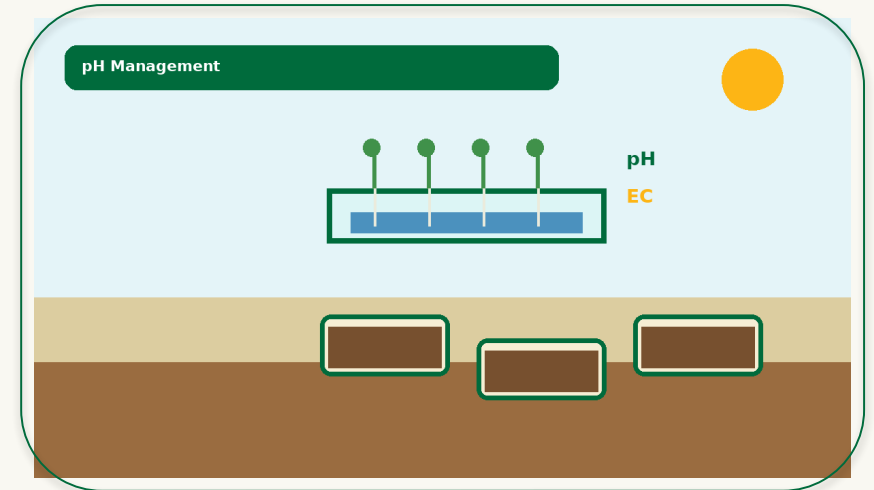
TEACHER PRACTICE

- Prepare a nutrient-solution management log and dilution exercise.

pH Management

Teacher technical knowledge, laboratory/field practice and student supervision

- Explain pH as a chemical property that influences nutrient availability and solution behavior.
- Use a calibrated pH meter or validated test method.
- Record pH, time, reservoir and crop/system identity.
- Interpret values against crop/system guidance rather than one universal target.
- Only authorized adults handle acid/base adjustment chemicals.



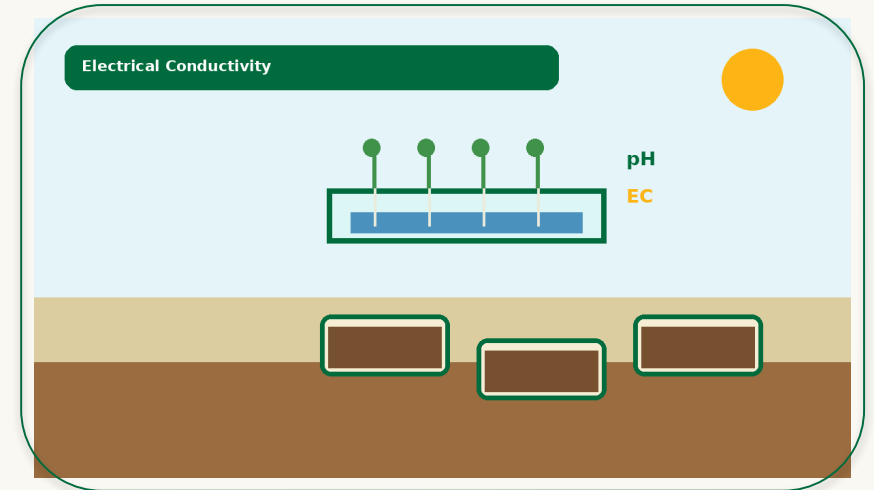
TEACHER PRACTICE

- Demonstrate calibration/checking and pH record interpretation.

Electrical Conductivity

Teacher technical knowledge, laboratory/field practice and student supervision

- Use EC as an indicator of the total dissolved ionic concentration of the nutrient solution.
- Record EC with units and instrument identification.
- Explain that EC does not reveal which specific nutrients are present.
- Interpret EC changes alongside water addition, nutrient dosing and crop uptake.
- Avoid correcting EC without considering source water and system management history.



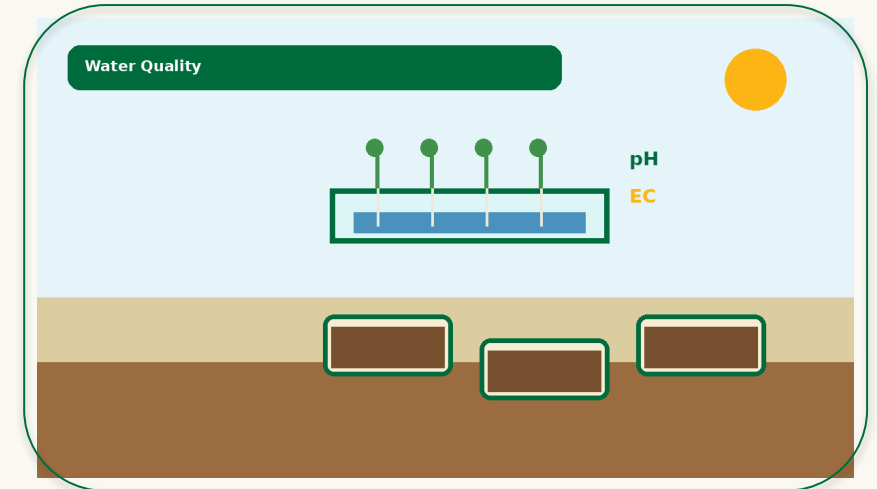
TEACHER PRACTICE

- Prepare an EC trend-analysis activity.

Water Quality

Teacher technical knowledge, laboratory/field practice and student supervision

- Assess source water using available information on pH, EC/salinity, suspended matter and other relevant quality indicators.
- Recognize that clear water can still contain dissolved salts or microbial contaminants.
- Document source changes and treatment steps.
- Match water-quality assessment to system sensitivity and crop requirements.
- Keep drinking-water and nutrient-system equipment separated where required.



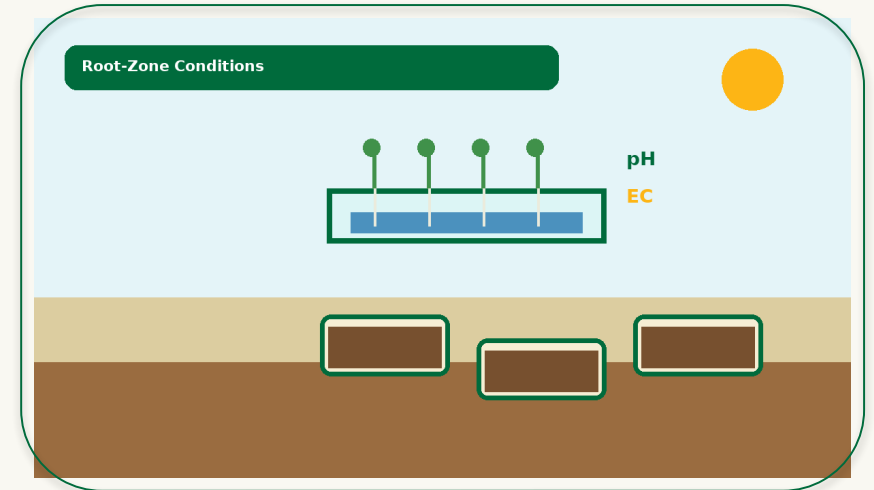
TEACHER PRACTICE

- Create a source-water suitability checklist.

Root-Zone Conditions

Teacher technical knowledge, laboratory/field practice and student supervision

- Monitor moisture/solution contact, oxygen availability, root temperature where measurable and visible root condition.
- Explain how poor aeration, excessive heat or interrupted water delivery can affect roots.
- Use safe viewing points rather than repeatedly disturbing plants.
- Record root observations with system measurements.
- Treat root discoloration as an observation requiring investigation, not an automatic diagnosis.



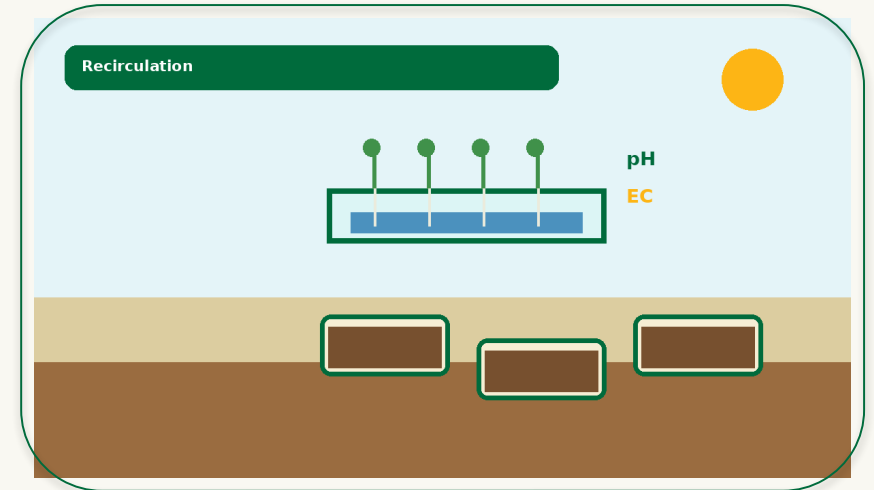
TEACHER PRACTICE

- Develop a root-zone monitoring protocol.

Recirculation

Teacher technical knowledge, laboratory/field practice and student supervision

- Trace solution from reservoir through crops and back to storage.
- Explain potential resource benefits and the increased importance of monitoring shared solution quality.
- Check return flow, reservoir level and pump operation.
- Recognize that pathogens or chemical imbalances can move through a shared loop.
- Maintain overflow and drainage safeguards.



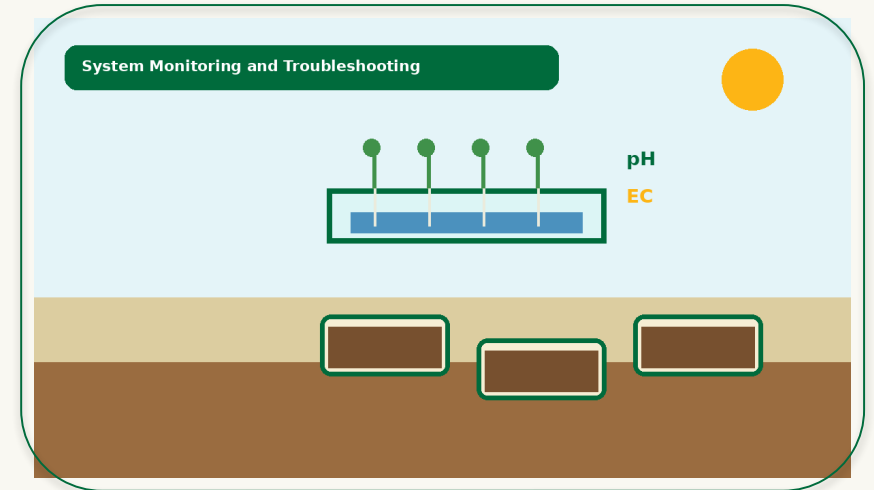
TEACHER PRACTICE

- Map the recirculation loop and identify control/monitoring points.

System Monitoring and Troubleshooting

Teacher technical knowledge, laboratory/field practice and student supervision

- Monitor pH, EC, water level, flow, temperature and crop/root observations as appropriate to the system.
- Use a structured troubleshooting sequence from symptom to measurement to likely cause.
- Change one management variable at a time where possible.
- Document corrective actions and verify system response afterward.
- Adults perform electrical, pump, chemical and pressurized repairs.



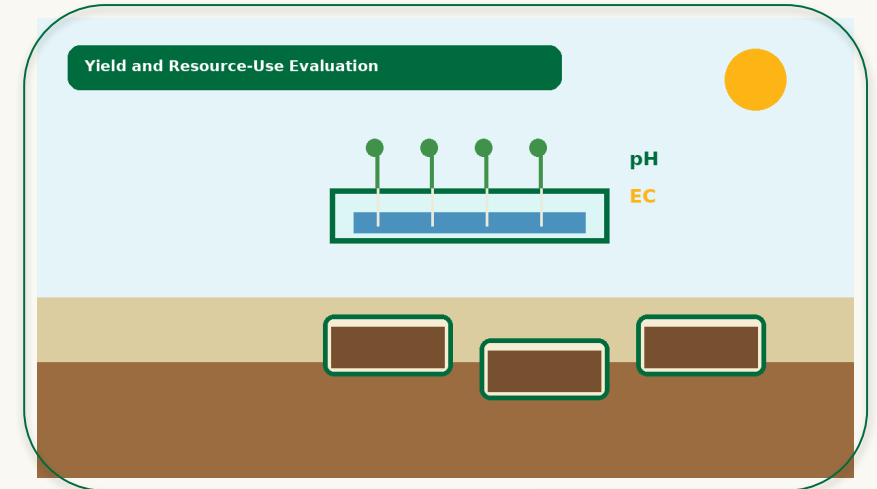
TEACHER PRACTICE

- Prepare a hydroponic troubleshooting decision tree.

Yield and Resource-Use Evaluation

Teacher technical knowledge, laboratory/field practice and student supervision

- Measure crop yield and quality from a defined system area or number of plants.
- Record water additions, nutrient inputs, energy use where measurable and labor/maintenance requirements.
- Calculate clearly defined resource-use indicators with units.
- Compare performance with a reference system cautiously.
- Evaluate yield alongside reliability, input demand and management complexity.



TEACHER PRACTICE

- Completion task: produce a multi-criteria soilless-system performance report.